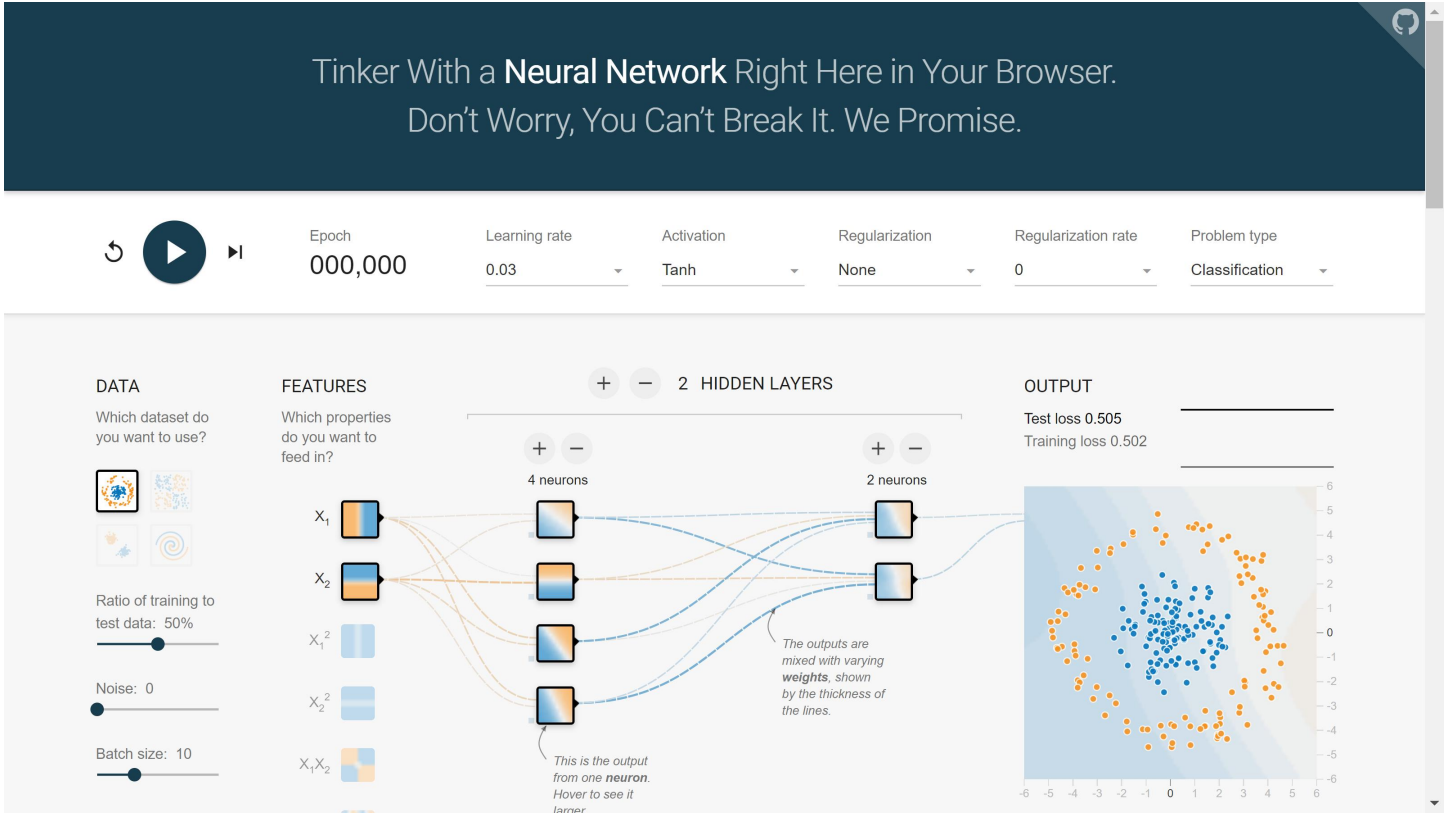
Network


Tinker With a Neural Network Right Here in Your Browser.
Don't Worry, You Can't Break It. We Promise.
Epoch
000,000
Learning rate
0.03
Activation
Tanh
Regularization
None
Regularization rate
0
Problem type
Classification
DATA
Which dataset do you want to use?
Ratio of training to test data: 50%
Noise: 0
Batch size: 10
FEATURES
Which properties do you want to feed in?
2 HIDDEN LAYERS
4 neurons
2 neurons
This is the output from one neuron. Hover to see it larger.
The outputs are mixed with varying weights, shown by the thickness of the lines.
OUTPUT
Test loss 0.505
Training loss 0.502

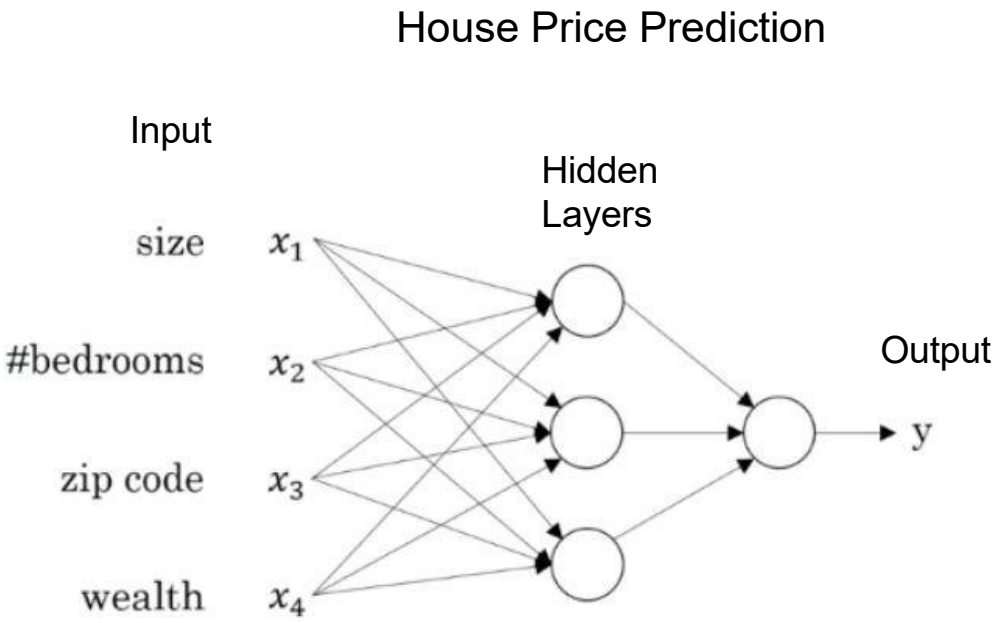
Outline

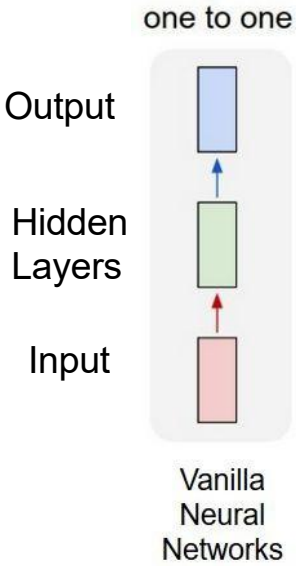
- **Deep Learning**
  - **RNN**
  - **CNN**
  - **Attention**
  - **Transformer**

RNNs

Some slides borrowed from Fei-Fei Li & Justin Johnson & Serena Yeung at Stanford.

Vanilla Neural Networks



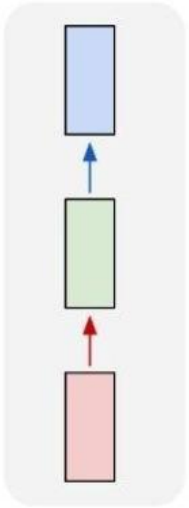


How to model sequences?

- Text Classification: Input Sequence -> Output label
- Translation: Input Sequence -> Output Sequence
- Image Captioning: Input image -> Output Sequence

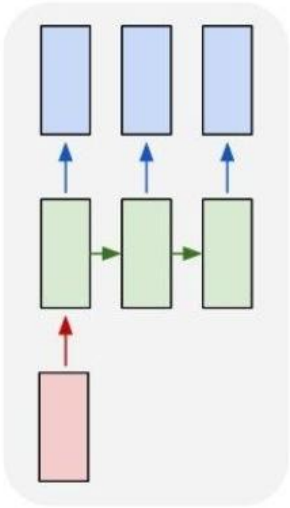
RNN- Recurrent Neural Networks

one to one



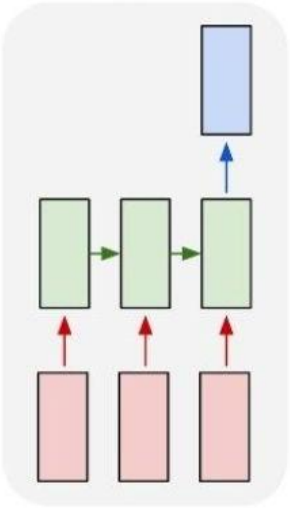
Vanilla Neural Networks

one to many



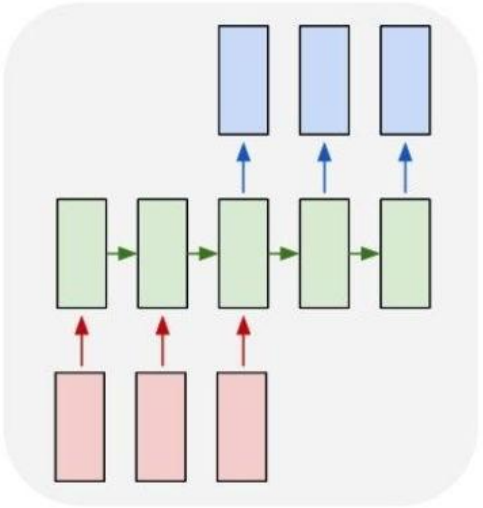
e.g.- Image Captioning

many to one



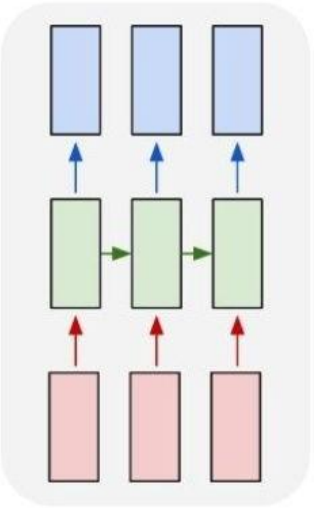
e.g.- Text Classification

many to many



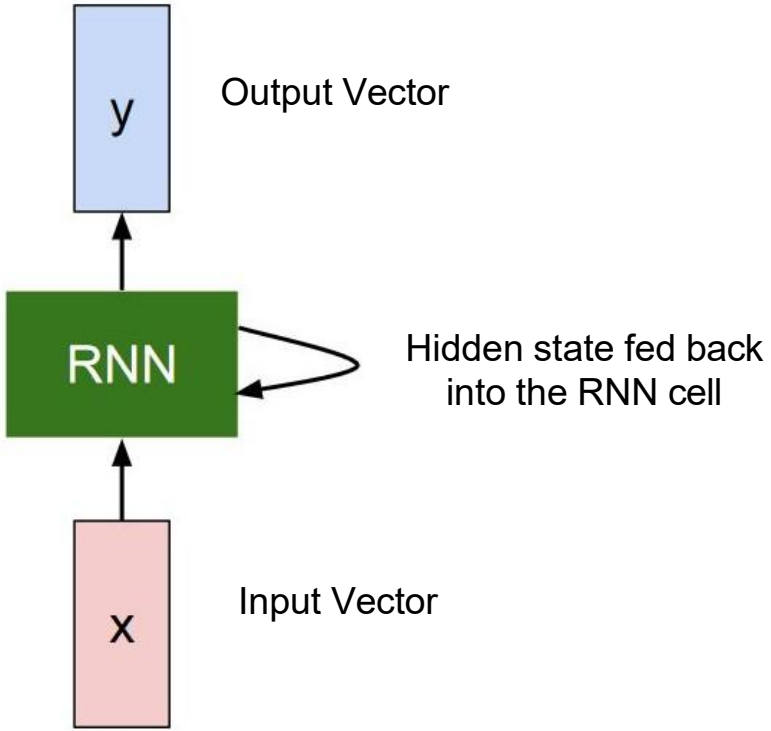
e.g.- Translation

many to many



e.g.- POS tagging

RNN- Representation



RNN- Recurrence Relation

The RNN cell consists of a hidden state that is updated whenever a new input is received. At every time step, this hidden state is fed back into the RNN cell.

$$h_t = f_W(h_{t-1}, x_t)$$

new state

some function with parameters W

old state

input vector at some time step

